

MARCOS JUAREZ, Argentina

WMO: 874670

Lat: 32.6797S

Lon: 62.1506W

Elev: 374

StdP: 14.50

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.8	32.7	21.9	16.6	39.6	24.7	19.0	41.2	33.5	51.8	28.7	52.1	2.6	50	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.6	94.0	75.2	91.6	74.0	89.2	73.0	79.2	89.1	77.4	87.1	75.7	84.8	9.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	140.1	85.6	74.6	131.3	83.2	73.0	124.1	81.2	43.2	89.5	41.3	87.3	39.6	85.1	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.4	23.4	20.2	24.0	99.0	2.8	2.4	21.9	100.7	20.3	102.1	18.7	103.5	16.6	105.2		
(5)			22.8	82.7	2.6	1.7	20.9	83.9	19.3	84.8	17.8	85.8	15.9	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	74.7	72.7	70.0	64.3	57.9	52.1	50.6	54.5	58.5	64.4	69.8	73.6		
	DBStd	10.52	4.93	5.60	5.97	6.72	6.91	6.66	7.01	8.03	7.11	6.37	5.78	5.32		
	HDD50	186	0	0	0	1	10	48	75	41	11	0	0	0		
	HDD65	1873	1	6	21	91	241	391	451	344	216	87	21	3		
	CDD50	5132	766	635	620	431	256	111	93	179	267	447	595	732		
	CDD65	1347	303	222	176	70	22	5	4	17	22	69	166	271		
	CDH74	13149	2960	1940	1569	607	139	19	33	203	313	803	1774	2789		
	CDH80	5107	1247	762	571	166	23	1	8	70	83	265	698	1213		
(14) Wind	WSAvg	7.7	6.4	6.1	6.3	7.0	6.5	7.1	7.7	9.0	9.5	9.7	8.8	8.1		
(15) Precipitation	PrecAvg	34.8	4.6	4.0	4.6	3.1	1.5	1.2	0.9	0.9	2.0	3.8	4.0	5.5		
	PrecMax	51.7	13.8	14.7	10.6	10.0	4.4	7.2	4.6	3.7	7.6	11.8	8.6	14.5		
	PrecMin	23.5	0.4	1.1	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.7	1.5		
	PrecStd	7.5	2.6	2.6	2.6	2.2	1.3	1.6	1.0	0.9	1.9	2.3	2.1	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.2	95.0	93.1	89.1	83.1	76.8	78.9	88.3	88.1	92.4	94.8	96.9		
		MCWB	78.1	78.4	75.3	73.8	70.0	67.7	66.1	67.9	69.1	71.9	72.6	76.4		
	2%	DB	92.7	91.2	89.4	84.7	77.7	70.0	71.4	79.2	82.0	86.9	91.1	93.3		
		MCWB	76.4	76.4	73.6	70.5	67.0	61.9	61.0	64.6	65.1	68.5	70.1	74.2		
	5%	DB	89.9	88.2	86.3	80.9	73.4	66.2	67.0	72.9	77.4	82.3	87.8	90.4		
		MCWB	75.2	75.3	72.3	67.6	64.5	59.6	58.0	60.8	62.1	66.0	68.4	72.6		
	10%	DB	87.1	84.9	82.9	77.1	69.4	63.2	63.1	68.4	73.3	78.1	84.1	87.0		
		MCWB	74.0	73.3	70.3	66.3	62.4	57.1	55.1	57.4	59.9	64.1	67.2	71.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	81.5	79.4	76.5	72.3	68.4	67.4	70.4	71.5	75.2	76.2	80.9		
		MCDB	91.4	91.5	89.7	85.4	79.2	74.5	75.9	83.7	84.1	88.2	89.4	91.8		
	2%	WB	79.0	78.9	76.2	73.1	69.0	64.8	63.6	66.5	67.9	71.2	73.0	77.6		
		MCDB	89.0	88.0	84.9	80.4	75.1	68.1	70.2	76.7	77.3	82.5	85.2	87.9		
	5%	WB	77.2	76.6	74.0	70.5	66.3	61.9	60.3	63.0	64.7	68.6	71.0	75.4		
		MCDB	86.8	85.1	82.9	77.1	71.7	64.6	64.3	70.4	73.8	77.9	82.3	85.7		
	10%	WB	75.4	74.7	72.0	68.2	63.5	58.9	57.3	59.5	61.8	66.3	69.0	73.2		
		MCDB	84.1	82.5	80.0	74.7	68.6	61.6	61.0	65.9	70.1	75.6	80.3	83.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.6	20.3	21.5	20.5	19.3	19.9	21.5	24.0	24.4	22.7	23.4	22.6		
		MCDBR	26.0	25.2	25.7	26.1	23.2	22.8	26.9	29.6	31.1	29.3	29.9	28.1		
		MCWBR	12.5	12.0	11.8	12.9	12.7	14.4	16.2	16.1	16.0	13.5	12.6	12.2		
	5% WB	MCDBR	22.5	21.2	21.6	20.6	19.1	16.0	20.9	25.9	25.5	24.3	25.2	23.3		
		MCWBR	12.4	11.4	11.2	11.3	11.6	12.0	13.9	15.1	15.0	13.3	12.7	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.408	0.390	0.376	0.373	0.350	0.333	0.342	0.393	0.452	0.416	0.396	0.397		
	taud		2.320	2.386	2.426	2.404	2.420	2.464	2.407	2.231	2.063	2.234	2.321	2.340		
	Ebn at Noon		295	294	287	269	259	256	259	257	258	283	297	300		
	Edh at Noon		43	39	35	32	28	25	28	38	50	45	43	43		
(44) All-Sky Solar Radiation	RadAvg		2287	1969	1667	1210	888	784	859	1156	1524	1819	2186	2324		
	RadStd		129	151	136	151	84	101	84	109	133	145	133	148		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	+0.90	+1.31	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	+0.41	+0.88	+0.99	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page